

WHAT IS THE CASE FOR PAID MATERNITY LEAVE?

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Abstract—We assess the case for generous government-funded maternity leave, focusing on a series of policy reforms in Norway that expanded paid leave from 18 to 35 weeks. We find the reforms do not crowd out unpaid leave and that mothers spend more time at home without a reduction in family income. The increased maternity leave has little effect on children's schooling, parental earnings and labor force participation, completed fertility, marriage, or divorce. The expansions, whose net costs amounted to 0.25% of GDP, have negative redistribution properties and imply a considerable increases in taxes at a cost to economic efficiency.

I. Introduction

THE past fifty years have seen a steady increase in the number of countries offering government-funded parental leave.¹ By 2013, all OECD countries except for the United States had federally funded programs, which generally offer high earnings replacement and employment protection to mothers who were working prior to giving birth. The even more striking trend has been the substantial increase in the duration of paid leave. As illustrated in figure 1a, the median number of weeks of paid leave to mothers among OECD countries was 14 in 1980 and had risen to 42 weeks by 2011. From a policy perspective, it is important to distinguish between the introduction of paid leave (and job protection) versus continual expansions to a program. Government-mandated leave programs involve large transfer payments to a sizable population, and the key question is whether benefits outweigh costs on both the extensive and intensive margins.

The main argument for paid leave is that there are important social goals or market failures addressed by the program. In the policy arena, proponents claim that paid leave could enhance child development by increasing parental time investment, promote gender equality through higher maternal employment, and encourage fertility and marriage by facilitating family and career compatibility. Paid leave could also serve as a transfer program with desirable redistribution effects. If there are no public benefits, however, paid leave can be justified only if it alleviates market failures. In particular, binding credit constraints could limit the intertemporal

substitution of maternal labor supply after birth or prevent families from investing optimally in children.

A comprehensive evaluation of the case for paid parental leave requires answers to at least four key questions. First, does paid leave increase available parental time with children, or does it simply crowd out unpaid leave? Second, what effect does paid leave have on a broad range of child, parent, and family outcomes? Third, how do any benefits compare relative to costs? And finally, are there any distributional effects?

We answer these questions in the context of Norway's parental leave program, focusing on parents' responses to the evolution of leave benefits. Seven major reforms have expanded the amount of leave mothers take without having a significant effect on fathers.² In 1977, Norway introduced eighteen weeks of paid parental leave and extended job protection from twelve weeks to one year. In the ensuing years, from 1987 to 1992, a series of policy reforms expanded paid leave from 18 weeks to 35 weeks (without changing the length of job protection). While prior work has documented an impact on children's high school graduation rates and future wages from the initial 1977 reform (Carneiro, Løken, & Salvanes, 2015), we investigate whether the subsequent generous expansions had any added effect on outcomes that could justify their additional costs. At a time when most countries are facing harsh budget realities, it is important to critically evaluate whether these fiscally expensive parental leave extensions had the desired policy effects.

Each of the reforms specified a birth cutoff date to determine eligibility for benefits. Parents of children born after the cutoff date were eligible for extra weeks of paid leave relative to parents with children born before the cutoff. We use a regression discontinuity (RD) estimator to obtain causal estimates of the effect of extra weeks of paid leave on a variety of outcomes. The idea is to compare families to the left and right of this cutoff, since they should be similar on average except for the extra allotment of paid leave from the government. By estimating the effect for all expansionary reforms, we can assess whether child, family, or social benefits continue to accrue as the generosity of the program increases, or whether additional weeks of paid parental leave simply represent transfer payments without measurable differences in outcomes.

Our first empirical result is that none of the reforms crowd out unpaid leave. Each reform increases the amount of time spent at home versus work by roughly the increased number of weeks allowed. Since income replacement was 100%

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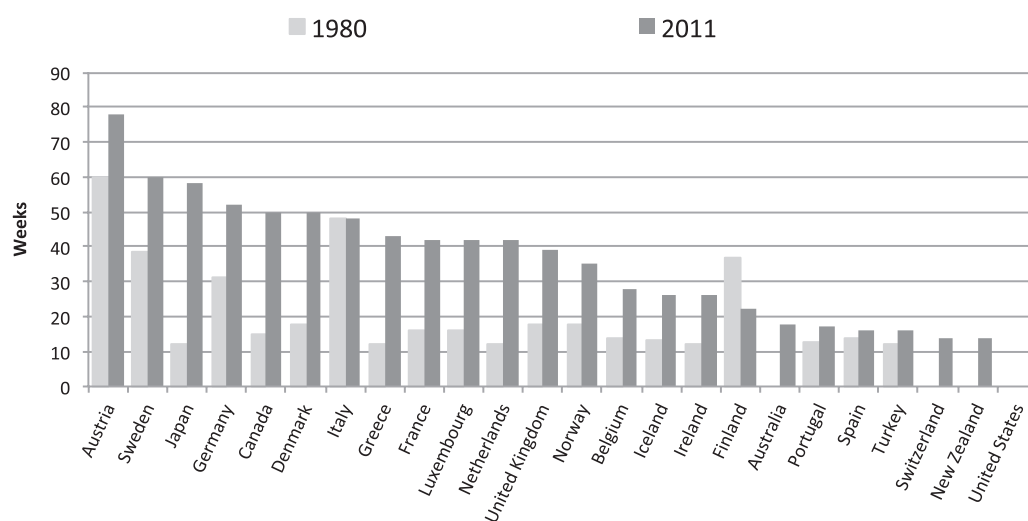
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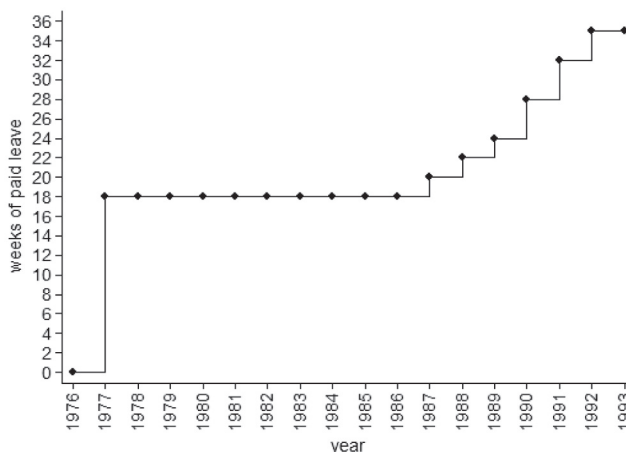
¹ Austria, the Czech Republic, Italy, and Poland introduced paid parental leave in the 1960s or earlier; Hungary, Spain, Sweden, Finland, France, and Norway in the 1970s; Iceland, Denmark, Portugal, Turkey, Greece, Germany, and New Zealand in the 1980s; Canada, the Netherlands, Japan, the Slovak Republic, Australia, Belgium, Ireland, and Luxembourg in the 1990s; and the United Kingdom in the 2000s.

² With the exception of a few weeks specifically reserved for mothers, these leave reforms allowed couples to divide up the weeks of leave as they saw fit. In practice, however, mothers took the entire amount of leave. Throughout this paper, we therefore refer to parental leave and maternity leave interchangeably.

FIGURE 1.—PAID PARENTAL LEAVE IN OECD COUNTRIES AND IN NORWAY OVER TIME



(a) Weeks of paid leave across OECD countries over time



(b) Expansions in paid leave in Norway over time

Data for panel a come from www.oecd.org/els/social/family/database.

for most women, the reforms caused an increase in mother's time spent at home after birth without a reduction in family income. There was almost no governmentally provided child care coverage for children under the age of 2 during this period (Havnes & Mogstad, 2011). As a result, the counterfactual to a mother's care is usually informal care, including relatives and nannies.³ Our setting allows for a clean estimate of the effect of parental time (relative to informal care arrangements) on child and family outcomes, as there are no income effects to worry about. The same is not true in other settings where income is only partially replaced or there is crowd-out of unpaid leave.

³ Note our reforms also occur before the introduction of cash-for-care benefits for children between the ages of 1 to 3 years old (introduced in 1998). There was only one other program (with no changes during our time period) for families with small children, the cash allowance program, which gives a direct, flat-rate cash benefit per child from birth to age 18.

Our second set of empirical results reveals that the expansions had little effect on a wide variety of outcomes other than mothers' time spent at home after birth. Not only is there no evidence that each expansion in isolation had economically significant effects, but this null result holds even if we cumulate our estimates across all expansions from 18 to 35 weeks. We find no effect of paid leave on children's academic achievement or graduation from high school. We also find no evidence of systematic changes in parental earnings or participation in the labor market in the short or long run, which means the leave expansions did nothing to gender equality on these dimensions. Finally, there is little evidence the expansions to parental leave affected completed fertility, marriage, or divorce.

Our third finding is that paid parental leave has negative redistribution properties. The program makes regressive transfers from ineligibles to eligibles, and among the group of eligible mothers, the payments increase with family

income. Since there is no crowd-out of unpaid leave, the extra leave benefits amount to a pure leisure transfer. Around 74% of all mothers were eligible for paid leave during our time period, since a requirement is that the mother works a certain amount prior to the child's birth. Mothers who are eligible and receive parental leave benefits are higher educated, are married to higher-educated men, and have substantially higher family income compared to ineligible mothers. The differences in income between ineligible and eligibles are even larger when comparing eligible mothers to the relatively poor men and women who never have children. Within the group of eligible mothers, the transfer payments were larger to women with higher prior earnings (since the program gave 100% income replacement up to a high threshold); not surprisingly, these women also reside in a family with higher total family income.

Finally, we investigate the financial costs of the extensions in paid parental leave. We find little, if any, impact on parents' future tax payments and benefit receipt. As a result, the large increases in public spending on parental leave imply a considerable increase in taxes at a cost to economic efficiency. We estimate that a one-week increase in paid leave costs taxpayers \$687 (in 2010 dollars) in program expenditures on average per birth, based on all the reforms we study.⁴ These costs add up: the initial 18-week reform cost an estimated total of \$12,354 per eligible birth, while the subsequent expansions from 18 to 35 weeks cost taxpayers \$11,668 per eligible birth. Stated differently, payments in 1992 when leave benefits were 35 weeks totaled over \$1 billion (in 2010 dollars), or approximately 0.5% of GDP.

Taken together, our findings suggest that the case for extensions to paid parental leave cannot rely on observable dimensions of public benefits or redistribution. Nor do paid leave extensions seem to alleviate binding credit constraints, which would limit intertemporal substitution of labor supply or prevent optimal investment in children: there is no evidence of changes in labor supply in the short or long run; there is no measurable effect on child development; and most transfer payments go to middle- or upper-class families, which are less likely to face binding credit constraints. In light of previous studies, one can argue whether a baseline level of paid parental leave or job protection is worth the cost. But the case for extended paid leave periods, which are prevalent in OECD countries, is weak. One would have to argue there are empirically important benefits not captured in the relatively rich set of outcome variables we look at. We discuss this possibility in more detail in the conclusion of the paper.

Our findings are related to a growing literature on paid parental leave introductions and extensions. Researchers have examined the effect of paid leave on parental

employment and earnings, child outcomes, fertility, and marriage.⁵ While these papers make important contributions to our understanding of parental leave, there is no consensus across studies. Consider the research most closely related to our study of paid leave extensions. In a series of papers, Baker and Milligan (2008a, 2008b, 2010) evaluate the extension of leave from 25 to 50 weeks in Canada, with an earnings replacement rate of 55%. They find that while it increased breastfeeding, it had little effect on child health and development up to age 3 and negative effects on verbal and self-awareness scores at ages 4 and 5. Rasmussen (2010) finds that a single expansion of leave in Denmark did not affect later educational outcomes, while Liu and Skans (2010) find that a single leave expansion in Sweden resulted in increased academic test scores for children of highly educated mothers but otherwise had no effect. Finally, Dustmann and Schönberg (2011) examine two paid leave and job protection expansions (from 2 to 6 months and from 6 to 10 months), as well as an increase in job protection from 18 to 36 months. These expansions provided a low level of earnings replacement (a flat amount equal to about 20% to 30% of pre-earnings, plus a means tested amount after 7 months). They find little effect of the first two expansions, but some negative effects for the increase in job protection without a corresponding rise in payments.

Our paper expands and clarifies this prior literature in several important ways. First, we are able to look at a broad range of outcomes in the same setting over time. Second, we provide direct evidence on the cost side of these policies and their distributional effects. Third, we examine a series of six expansions. Not only does this give us multiple quasi-experiments, it also helps determine whether modest expansions are worthwhile, even if further, larger expansions are not. A comparison between benefits and costs of marginal expansions is key to determining the optimal duration of paid parental leave. Fourth, each of the reforms we study extended paid leave without any changes in job protection, which helps isolate the effect of paid leave from employment protection. Fifth, take-up is essentially 100% in terms of participation and utilization of all weeks of leave, in contrast to other countries with less generous leave programs. Finally, we are able to hone in on the counterfactual to paid leave because there is no evidence of crowd out of unpaid leave and there was 100% income replacement. This helps us interpret the expansions to paid parental leave as increases in the amount of time spent at home versus work, holding family income constant.

Our paper is also related to a large literature on maternal employment and child outcomes (see Blau & Currie, 2006, for a review). That literature is largely inconclusive, with much of the work suffering from endogeneity issues

⁴ Leave benefits are taxable at the individual level, but tax revenue is also lost due to the forgone income of mothers on leave. In our setting where there is no crowd-out of unpaid leave and 100% income replacement, these two tax effects roughly cancel each other out, so the after-tax costs of the program are approximately equal to program expenditures.

⁵ See, for example, Baker and Milligan (2008a, 2008b, 2010), Carneiro et al. (2015), Dustmann and Schönberg (2011), Lalive et al. (2014), Lalive and Zweimuller (2009), Liu and Skans (2010), Rasmussen (2010), Rossin (2011), Ruhm (2000), Schönberg and Ludsteck (2014), and Waldfogel, Higuchi, and Abe (1999).

related to which women choose to work (Gregg & Waldfogel, 2005; Tanaka, 2005; Gregg et al., 2005). Our setting is attractive to study the effect of maternal employment for two reasons. The first is that paid leave did not change the family's income, only mother's weeks of work. The second reason is that there are no concerns with sample selection as take-up is 100%; our estimates are therefore internally valid for working women whose children are born near the reform dates between 1987 and 1992. Our findings suggest that maternal employment between the ages of roughly 4 to 9 months has little effect on a child's later outcomes (as well as no effect on a couple's labor market participation, fertility, or family stability), even when family income remains constant and the alternative to a mother's time is informal care arrangements.

The remainder of the paper proceeds as follows. The next section provides background on paid parental leave in Norway and our data. Section III briefly discusses our regression discontinuity design and threats to identification. In section IV, we present our main results, followed by a comparison of costs versus benefits and distributional effects in sections V and VI. Section VII concludes.

II. Background and Data

A. Background

Governmental paid parental leave has a long history in Norway.⁶ In 1977, parents were granted 18 weeks of paid leave and job protection for paid and unpaid leave up to one year.⁷ During the late 1980s and early 1990s, the paid leave period gradually expanded, so that by 1992, there was a maximum of 35 weeks of paid leave but no increase in job protection. Figure 1b gives an overview of the introduction of paid parental leave in 1977 and the yearly expansions from 1987 to 1992.

The parental leave mandates provide 100% income replacement up to a generous earnings threshold. The thresholds are nonbinding for most mothers, and when they are exceeded, most public and private employers top up benefits so that forgone earnings are fully replaced.⁸ The firm is not allowed to dismiss workers for taking leave, and parents have the right to return to a comparable job. Apart from a few weeks reserved for the mother, parents could share 35 weeks between them as desired before 1993. Up to that point, the program was de facto a maternity leave program, as very few fathers took any leave.⁹

In our analysis, we focus on the six expansions in paid leave between 1987 and 1992. There are several reasons

for this choice. First, this allows us to analyze long-term outcomes, such as children's high school graduation rates and parents' long-term labor force attachment. Second, this period gives us a total increase of 17 weeks of paid leave, providing a natural comparison to the initial 18 weeks introduced in 1977. Finally, this is the period before the reforms started to set aside specific quotas for fathers, which would add a new layer of complexity. While virtually all eligible mothers take leave, not all eligible fathers do, even after the introduction of the father quotas.

The various reforms determined benefits based on the birthdate of the child. For the 1987 reform, parents with children born after May 1, 1987, were eligible for twenty weeks of paid parental leave, while parents whose children were born before were eligible for just eighteen weeks. Subsequent expansions were introduced with the following date cutoffs: July 1, 1988: two additional weeks; April 1, 1989: two additional weeks; May 1, 1990: four additional weeks; July 1, 1991: four additional weeks; and April 1, 1992: three additional weeks.¹⁰

In addition to the birth cutoffs, paid parental leave benefits were contingent on the mother's working at least six of the previous ten months before birth. Earnings in the prior ten months needed to exceed the "substantial gainful activity" threshold (approximately \$12,500 in the year 2010).

The parental leave system is universal, simple, and well known (including details about eligibility, benefit amounts, and the application process). To apply for parental leave benefits, parents must inform their employers and submit a joint application to a Social Security Administration field office at least six weeks before the pregnancy due date.

B. Data

Our analysis employs several data sources that we can link through unique identifiers for each individual. This rich longitudinal database covers every resident starting in 1967 and contains individual demographic information and unique identifiers that allow us to match spouses and parents to their children. The data on parental leave come from social security registers that contain complete records for all individuals from 1992 onward. One advantage of our data is that we can study a wide variety of parent, child, and family outcomes.

For mothers and fathers, we focus on outcomes related to annual earnings and transfers, as the Norwegian data do not include a measure of hours. Income is measured as total pension-qualifying earnings and transfers reported in the tax registry. This individual-specific measure includes labor earnings and all taxable transfers, including sickness benefits, unemployment benefits, and parental leave payments. We construct a measure of whether mothers work after birth

⁶ Our discussion of parental leave in Norway builds on Carneiro et al. (2015), Dahl et al. (2014), and Rege and Solli (2010).

⁷ Prior to this, parents had 12 weeks of job protection (since 1956) but no paid leave.

⁸ In 2010, benefits were capped at earnings of approximately \$75,000. Only 7% of mothers earned more than this threshold.

⁹ Starting in 1993, father quotas have been added in subsequent reforms, mostly on top of existing parental leave. See Dahl et al. (2014) for an analysis of the 1993 paternity leave quota.

¹⁰ Beginning with the 1989 reform, parents were given the option of taking 80% earnings replacement and 20% more weeks of paid leave. Since job protection extends for a year, at the margin it should be the same to take extra weeks at 80% coverage or 100% coverage and additional weeks as unpaid leave.

based on whether they have positive labor earnings. We also look directly at earnings after the reforms. For both mothers and fathers, we construct two long-term outcomes: total years of employment up to fourteen years after the reform and the annuity of income up to fourteen years after each reform.

The outcomes we study for children are the written exam taken at the end of junior high school and dropout rates from high school. The written exam is important for determining placement into high school and must be taken by all students. The exam subject rotates and can be in math, Norwegian, or English. High school dropouts are defined as students not obtaining a three-year high school diploma by age 20. Due to data availability, we do not have dropout status for the children affected by the 1992 reform.

For family outcomes, we examine combined family income, marriage patterns, and fertility. We measure long-term marital stability by observing whether a couple is divorced fourteen years after the reform, conditional on being married the year before the reform. We also look at whether a couple is married fourteen years later, given that they were not married the year before the reform. Finally, we look at completed fertility, defined as the cumulative number of children born to a mother fourteen years after the reform.

One of the key advantages of our data set is that we can address the cost side of parental leave, as well as its distributional effects. We can measure the amount of taxes a couple pays in the fourteen years after the program to see if the program is self-funding in the long run due to increased female labor force participation or earnings. We can also measure the future benefits from other programs a family receives in the long run, including the universal child allowance; the family allowance, which is targeted to single parents; and disability insurance. With these variables, we can compare the annuity of costs to the annuity of benefits.¹¹

Since we do not observe hours, we cannot directly observe which parents are eligible for paid leave according to the rules described in the previous section. We therefore predict eligibility using labor earnings the year before birth. We define eligibility for mothers as having earnings above the substantial gainful activity level, described in the previous section, in the year prior to birth. Eligibility increases over time as more women enter the labor market. For example, in 1987, about 70% of women are eligible, but by 1992, 76% of women are eligible. We limit our estimation sample to mothers who are predicted to be eligible.

There is a trade-off between using too strict an earnings requirement and excluding parents from our sample who were in fact eligible, and using a less strict earnings requirement and including parents who actually were ineligible. While including ineligible parents may increase the residual variation and thus the standard errors, excluding eligible

parents may affect the external validity of our results.¹² By using a fairly weak earnings requirement in the prediction of eligibility, we assign more weight to the generalizability of our results.

The final variable needed for our analysis is take-up of parental leave. The exact amount of paid leave taken by each parent and the amount of the benefit is directly observed from 1992 on. For prior years, we assume that paid leave has a take-up rate of 100% for mothers (with no take-up for fathers). This assumption is supported both by survey evidence (Carneiro et al., 2015) and the fact that observed take-up for the 1992 reform is close to universal for mothers and close to 0 for fathers. To calculate the amount of unpaid leave, we follow Carneiro et al. (2015) and impute days of unpaid leave using information on yearly earnings.¹³ Later in the paper, we verify this method by comparing the imputed estimates for 1992 to the actual data on leave taking.

Appendix table A1 provides summary statistics for the variables defined in this section.

III. Identification

A. Regression Discontinuity Design

While we analyze six different expansions in paid leave, in each case we use a similar identification strategy. To make things concrete, we focus the discussion on the 1992 reform. The discontinuity we exploit arises from the reform being contingent on the birthdate of the child. Parents of children born after April 1, 1992, were entitled to 35 weeks of parental leave, while those with a child born before this cutoff could receive only 32 weeks. The reduced-form model for our RD design can be implemented by the following equation,¹⁴

$$y = \alpha + 1[t \geq c](g_l(t - c) + \lambda) + 1[t < c]g_r(c - t) + e, \quad (1)$$

where y is the relevant outcome variable, t is the birthdate of the child, c is the cutoff date, e is an error term, and g_l and g_r are unknown functions. The coefficient λ is the intention-to-treat (ITT) effect of the reform on outcomes, where the reform specifies the increase in the maximum number of weeks of paid leave a mother can take. To get the average effect of the extra weeks of paid leave, λ needs to be scaled by the jump in the take-up of parental leave at around the cutoff from a first-stage RD regression. For 1992, when actual

¹² Note, however, that as long as eligibility cannot be manipulated, the internal validity of the RD estimates is unaffected by the exclusion of ineligible.

¹³ Specifically, we impute prebirth monthly income by dividing earnings the year before the reform by 12. Then we calculate total earnings in the year of the reform and the following three years after, and divide this by prereform monthly income. This yields a measure of the number of days of unpaid leave during the first 36 months after birth. For this imputation to work, the assumption is that prereform hourly wages are a good approximation for maternal potential postbirth hourly wages (the hourly wage rate a mother would have received had she not gone on unpaid leave), adjusted for inflation.

¹⁴ See Imbens and Lemieux (2008) and Lee and Lemieux (2010) for details on the implementation and assessment of RD designs.

¹¹ See Blundell, Graber, and Mogstad (2012) for a description of the Norwegian tax-transfer system. For all annuities, we use an annual discount rate of 2.3%.

take-up is observed, we can estimate this first stage. For earlier years, this is not possible. However, since take-up of parental leave benefits for mothers is close to 100% in terms of participation and utilization of all weeks of leave in 1992 and survey data suggest the same holds true from the beginning of the program (Carneiro et al., 2015), the ITT estimate should be close to the average effect. For the remainder of the paper, therefore, we focus on the ITT estimates.

B. Threats to Identification

Strategic timing of births. The validity of our RD design requires that individuals cannot manipulate the assignment variable, which is the child's birthdate. If date of birth cannot be timed in response to the reform, the aggregate distribution of the assignment variable should be continuous around the cutoff date.

There is little opportunity to strategically time conception, as the implementation dates for the reforms have been announced less than nine months in advance. The national budgets that propose the parental leave reforms are generally introduced in the fall and passed by parliament in December of the same year. Therefore, mothers giving birth close to the cutoffs, which occurred in the spring and summer, were already pregnant before the announcement of the reform. Furthermore, as discussed in section IIA, the month of implementation varied from reform to reform, so the exact timing would be hard to predict.

While strategic timing of conception is unlikely, it is still possible that mothers with due dates close to the cutoff date could postpone induced births and planned cesarean sections. In contrast to current birth practices in the United States, the vast majority of births in Norway during the time of our reforms were spontaneous vaginal deliveries. In 1993, the C-section rate was 12% (with 59% of these being emergency operations), and only 12% of vaginal deliveries were induced.¹⁵ Therefore, this type of manipulation is less likely than it would be in the U.S. setting (Dickert-Conlin & Chandra, 1999).

Appendix figure A1 graphically confirms there is no systematic effect from any of the reforms on the distribution of fertility around the cutoffs. While there are some seasonal patterns in the number of births, the reforms in general show no systematic evidence that births are strategically delayed. Nonetheless, to avoid the possibility that some births in our sample are strategically delayed, we report robustness results that exclude the week immediately before and the week immediately following the cutoff date. As we will show, excluding these observations does not materially affect our findings.

¹⁵These statistics are reported by Folkehelseinstituttet (<http://mfrnesstar.uib.no/mfr>). By comparison, the United States had C-section and induction rates of, respectively, 33% and 23% in 2011 (National Vital Statistics Reports, Births: Final Data for 2011, vol. 62, no. 1, June 28, 2013).

Eligibility. Another threat to our identification strategy is that the announcement of the reform could cause a change in eligibility around the cutoff date. If mothers could manipulate eligibility precisely by increasing their hours worked, then restricting the sample to eligibles could bias the estimates. As explained above, we predict eligibility based on annual earnings in the year prior to childbirth. It is possible that mothers' earnings could respond to the announcement of the reform. Recall, however, that predicted eligibility of mothers who have a child in the window surrounding the reform is based on annual earnings in the prior year. As the reforms were usually announced in December, the mother has only one month in which to increase her earnings enough to become eligible by our definition. Given this short time frame, there is limited scope for mothers to manipulate predicted eligibility status precisely, which supports the internal validity of our estimates for working women whose children are born near the reform dates.

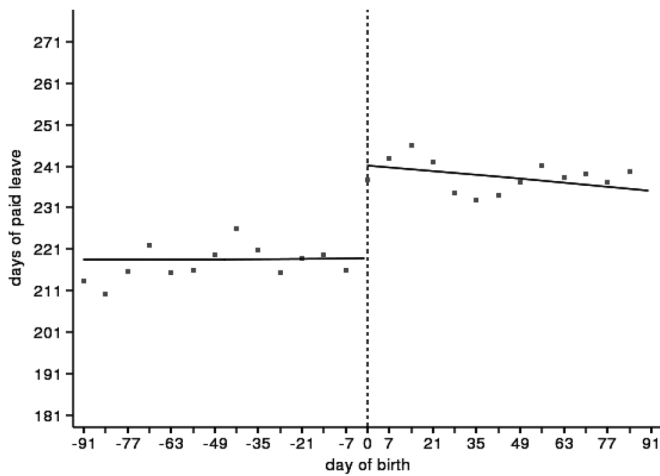
Appendix figure A2 graphically illustrates that there is no measurable change in predicted eligibility of mothers around the cutoff dates, except for the 1991 reform. While there is some seasonal variation in earnings, and thus in predicted eligibility, there is generally no jump in the fraction of predicted eligible mothers around the discontinuities, a finding confirmed with RD regressions in appendix table A2. For 1991, the only reform year with a statistically significant eligibility change, the estimate is actually negative, which is the opposite direction from what would be expected if mothers were manipulating their eligibility.

Covariate balance. If families time date of birth or change eligibility status in response to the reform, then we would expect to see changes in the distribution of predetermined characteristics of the parents around the reform dates. Appendix table A2 shows that there is little evidence to support this. Out of the 24 estimates, 3 are significant at the 10% level or less, which is not much more than would be expected by chance. Using an F -test, we cannot reject the hypothesis that all 24 coefficients are jointly equal to 0 ($F = 1.32, p\text{-value} = .14$). In addition, the point estimates are small in magnitude and our RD estimates barely move when we include these characteristics in the regressions.

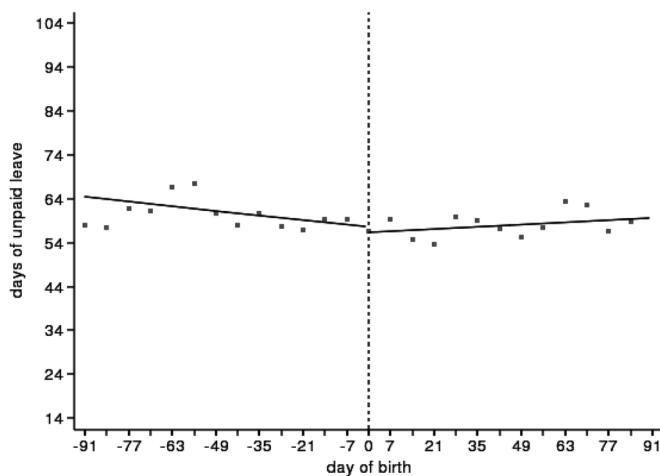
IV. Results

A key advantage of RD is that results can be presented graphically, which makes identifying the estimates more transparent. However, given the large number of outcomes we analyze, we cannot feasibly show graphs for every reform. Therefore, we first present results graphically for the 1992 reform in the paper. Results for the other expansions in parental leave are presented in table form, with graphs relegated to appendix figures A3 to A14.

FIGURE 2.—MATERNITY LEAVE, 1992 REFORM



(a) Paid leave, 1992 reform



(b) Unpaid leave, 1992 reform

Each observation is the average days of paid or unpaid leave for mothers in a one-week bin based on the birth date of the child. The dashed vertical lines denote the reform cutoff date, which has been normalized to 0. The solid lines are estimated using a linear regression based on daily individual-level data using triangular weights. The scales of the y-axes are $\pm .5$ standard deviations of the mean outcome.

A. First Stage

The 1992 reform increased the number of paid weeks of parental leave by three, changing the entitlement from 32 to 35 weeks. Figure 2b displays the number of days of paid leave taken by mothers in a window surrounding the reform. The graph includes births three months before and three months after the reform. In the graph, the birth date of the child has been normalized with the cutoff date of April 1, 1992, being labeled as 0. Each point on the graph is the average number of paid leave days taken by mothers over a one-week birth interval. There is a sharp jump in the number of days of paid leave at the birth cutoff date.

To obtain an RD estimate, we adopt the following specification for this first-stage regression and for subsequent reduced-form regressions. We use daily data, include linear trends in birth day on each side of the discontinuity, and

TABLE 1.—CROWD-OUT OF UNPAID LEAVE, ALL REFORMS

Reform	RD Estimate	
	Unpaid Leave (days)	Observations
1987	-2.041 (4.059) [102.14]	17,580
1988	1.341 (3.658) [95.06]	19,310
1989	-3.920 (3.473) [80.82]	20,093
1990	0.039 (2.991) [61.82]	21,508
1991	-2.726 (2.721) [56.49]	21,717
1992	-1.124 (2.564) [59.13]	21,838

All coefficients are estimated using a linear RD model and triangular weights using children born three months before and three months after the reform. Control variables include father's and mother's years of education and age at birth, marital status, and county of residence the year prior to birth, as well as an indicator of the child's gender. Unpaid leave measured in days. Standard errors are reported in parentheses and comparison means in brackets. * $p < 0.1$, ** $p < 0.05$.

employ triangular weights. To gain precision, we also include (predetermined) control variables for mother's and father's years of education, mother's and father's age and age squared, parents' county of residence, marital status prior to birth, and gender of the child. The estimated jump in this first-stage regression is 22.7 days ($SE = 2.7$), which corresponds closely to the three-week increase stipulated by the 1992 reform. This estimate confirms survey evidence suggesting 100% take-up by mothers of the entire amount of paid leave benefits.

As mentioned previously, we observe actual days of paid leave starting in 1992. In the results that follow, we therefore focus on reduced-form estimates since we lack a first stage for earlier reforms. However, since take-up is likely to have been close to 100%, these ITT estimates should be very close to the average treatment effects in magnitude.

Crowd-out of unpaid leave. An important question is whether an increase in the number of paid leave days reduces the number of unpaid leave days. If so, then the intention of the reforms to increase the amount of time parents spend with children would be only partially achieved. Moreover, if the unpaid leave decreases as paid leave increases, there would be important income effects associated with the introduction of the reforms, as total income (labor earnings plus leave benefits) would rise after the reform.

Figure 3b graphs the fraction of mothers taking unpaid leave based on the birth date of their child. There is no evidence of any change in unpaid leave as a result of the reform. Table 1 presents RD estimates for the 1992 reform, as well as the other expansion. All of the estimates are close to 0 and statistically insignificant. Unlike much of the previous research, we are able to hone in on the counterfactual

to paid leave because there is no evidence of crowd-out of unpaid leave, and there is 100% income replacement. This helps us interpret the expansions to paid parental leave as increases in the amount of time spent at home versus work, holding family income constant. Since there was almost no child care coverage for children under the age of 2 during this period (Havnes & Mogstad, 2011), the counterfactual to parental care is usually informal care, including relatives and nannies.

The fact that income replacement was 100% and there is no crowd-out of unpaid leave makes our results easier to interpret compared to prior research in other countries. Indeed, one possible explanation for why much of the previous literature either finds no effect or a negative effect of paid leave on child development is that there could be offsetting effects from a mother's increased time at home versus a reduction in family income. This is because in many other countries, the replacement rate for paid leave is substantially less than 100%, so if there is incomplete crowd-out of unpaid leave, total family income could drop.

Child achievement and schooling. One of the main arguments made for paid parental leave, and maternity leave in particular, is that it will have a positive impact on child development. The logic is that by reducing the amount of time spent at work, parents will have more time to invest in their children. Whether this is true is controversial. The research on how maternal employment affects children's outcomes is inconclusive, in part because it is difficult to control for the self-selection of mothers who choose to work and in part because shocks to employment usually coincide with large changes to income, which could also affect child development.

The Norwegian parental leave setting is ideal to study how maternal employment in the first year of life affects later child outcomes, as it does not suffer from the same deficiencies as most of the prior studies. First, since take-up is essentially 100% and the RD design generates variation, which is as good as random around the cutoff, there is no self-selection into which eligible mothers take more versus less paid leave after a reform. Of course, the estimates are only internally valid for the group of eligible mothers, but this is an important and sizable group (approximately three-fourths of all mothers). Second, the previous section revealed there is no crowd-out of unpaid leave, which means the reforms changed the amount of time mothers spent working but did not affect family resources in the short run. As we show in the next section, the reforms did not significantly change family resources (including income from all sources, and not just the mother's labor income) in the long run either. As a result, we have an exogenous change to maternal labor supply during the first year after birth, holding family income fixed.

We test for effects on two long-term outcomes related to a child's long-term development: written exams taken at the end of middle school (ninth grade) and whether the child

TABLE 2.—CHILD OUTCOMES, ALL REFORMS

Reform	RD Estimates		Observations	
	Ninth-Grade Exams	High School Dropout	Exam	Dropout
1987	−0.036 (0.033) [3.54]	−0.003 (0.014) [0.35]	16,650	17,580
1988	−0.025 (0.031) [3.51]	0.015 (0.014) [0.35]	18,290	19,310
1989	0.038 (0.030) [3.52]	0.007 (0.013) [0.33]	19,200	20,093
1990	−0.067** (0.029) [3.52]	−0.005 (0.013) [0.33]	20,056	21,508
1991	−0.000 (0.029) [3.52]	0.003 (0.013) [0.33]	20,459	21,717
1992	−0.010 (0.029) [3.60]	.	20,427	.

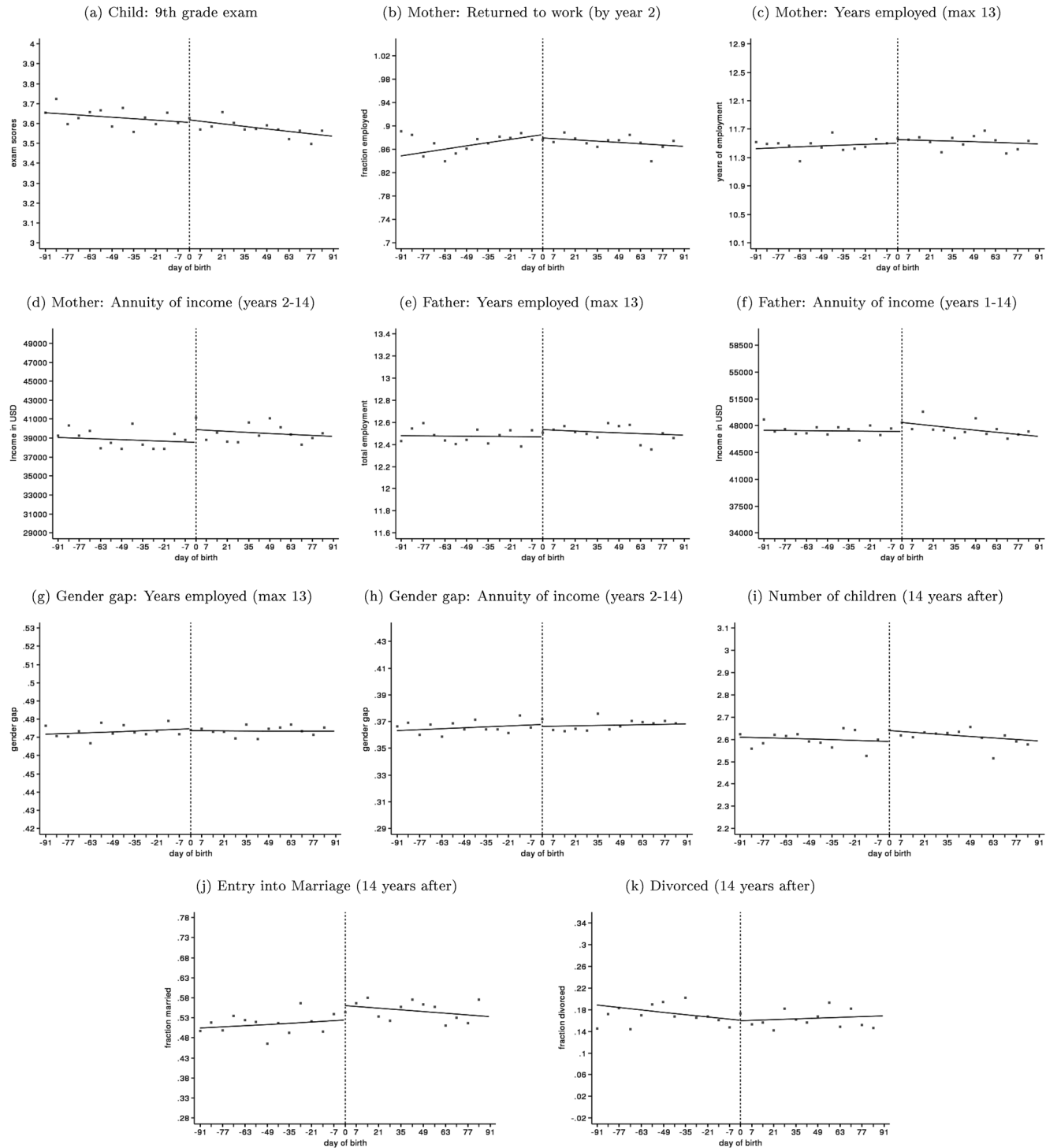
See notes to table 1 for the RD regression specification. Standard errors are reported in parentheses and comparison means in brackets. * $p < 0.1$, ** $p < 0.05$.

graduates from high school. The exam taken at the end of middle school is important for placement into high school. The scores range from 1 to 6, with a standard deviation approximately equal to 1. Figure 3a plots the average exam grade based on a child's birth date in a window surrounding the 1992 reform. Although exam scores are somewhat higher for children born earlier in the year, there is no jump in average scores at the reform cutoff date. Looking at table 2, there is similarly little evidence that expansions in paid leave affect exam scores for the other reforms. The effect sizes are close to 0, and only for the 1990 reform is the estimate significant. But the sign is negative, which, if taken literally, suggests that increased parental leave harms a child academically.

Our next child outcome is whether a child drops out of high school. Because children born in 1992 have not yet had a chance to graduate from high school by the end of our data sample, we cannot show a graph for this year. But we can report results for the other five reforms, which we do in column 2 of table 2. The RD estimates are close to 0, indicating no effect of paid leave on high school graduation. These last results stand in stark contrast to those of Carneiro et al. (2015), who find that the 1977 introduction of eighteen weeks of paid leave and extension of job protection from twelve weeks to one year decreased high school dropout rates by a statistically significant 2.7 percentage points.

Labor market outcomes. Another primary motivation for government leave programs is the claim that they will ease the transition of mothers back to work after the birth of a child and thereby promote increased attachment to the labor market over the long run. We graphically examine these claims for the 1992 reform in figures 3b to 3d. There is

FIGURE 3.—CHILD, MOTHER, FATHER, AND FAMILY OUTCOMES, 1992 REFORM



Each observation is the average of the relevant outcome in a one-week bin based on the birth date of the child. The dashed vertical lines denote the reform cutoff date, which has been normalized to 0. The solid lines are estimated using a linear regression based on daily individual-level data with triangular weights. The scales of the y-axes are $\pm .5$ standard deviations of the mean outcome.

no visual evidence that mothers have an increased rate of returning to work two years after the birth of their child. There is also little visual evidence of increased labor market attachment in the long run. Total years of employment and the annuity of income two to fourteen years after a child's

birth are largely unaffected by the three-week increase in paid leave. The results shown graphically for 1992 are not unique to that reform. Table 3 looks at the same outcomes for all six reforms. While a few of the coefficients are significant, the magnitudes are relatively small.

TABLE 3.—LABOR MARKET OUTCOMES OF MOTHERS AND FATHERS, ALL REFORMS

Reform	RD Estimates					Observations
	Mother: Returned to Work (by year 2)	Mother: Years Employed (Maximum 13)	Mother: Annuity of Income (years 2–14)	Father: Years Employed (Maximum 13)	Father: Annuity of Income (years 2–14)	
1987	−0.002 (0.013) [0.76]	−0.077 (0.107) [10.83]	755 (517) [31,891]	−0.015 (0.061) [12.38]	478 (941) [62,151]	17,580
1988	0.013 (0.012) [0.80]	0.020 (0.100) [10.91]	21 (493) [32,849]	−0.034 (0.061) [12.36]	−729 (874) [63,236]	19,310
1989	0.020* (0.011) [0.84]	−0.000 (0.090) [11.14]	−343 (629) [34,796]	−0.020 (0.053) [12.41]	1,474 (1,009) [64,478]	20,093
1990	0.009 (0.010) [0.85]	0.168** (0.085) [11.30]	708 (542) [36,252]	0.016 (0.052) [12.42]	1,307 (994) [66,477]	21,508
1991	0.014 (0.010) [0.86]	0.150* (0.084) [11.39]	770 (512) [37,384]	−0.003 (0.052) [12.45]	65 (976) [68,460]	21,717
1992	−0.007 (0.009) [0.87]	0.024 (0.078) [11.50]	877* (519) [39,229]	0.067 (0.050) [12.49]	1,060 (1,098) [70,545]	21,838

See notes to table 1 for the RD regression specification. Annuities are measured in year 2010 dollars. Standard errors are reported in parentheses and comparison means in brackets. * $p < 0.1$, ** $p < 0.05$.

TABLE 4.—LABOR MARKET GENDER GAPS, FERTILITY, MARRIAGE, AND DIVORCE, ALL REFORMS

Reform	RD Estimates					Observations		
	Gender Gap: Years Employed (Maximum 13)	Gender Gap: Annuity of Income (years 2–14)	Number of Children (14 years after)	Entry into Marriage (14 years after)	Divorced (14 years after)	Gender Gaps, Number of Children	Entry into Marriage	Divorced
1987	−0.004 (0.004) [0.45]	0.002 (0.005) [0.35]	−0.030 (0.029) [2.59]	−0.015 (0.023) [0.60]	0.012 (0.015) [0.14]	17,580	7,761	9,819
1988	0.002 (0.004) [0.46]	0.002 (0.005) [0.35]	0.017 (0.028) [2.60]	0.024 (0.022) [0.59]	−0.009 (0.016) [0.16]	19,310	9,346	9,964
1989	−0.001 (0.003) [0.46]	−0.003 (0.004) [0.36]	−0.016 (0.026) [2.60]	0.021 (0.022) [0.56]	0.022 (0.014) [0.16]	20,093	9,414	10,679
1990	0.006* (0.003) [0.47]	0.002 (0.004) [0.36]	−0.026 (0.026) [2.62]	−0.011 (0.021) [0.55]	−0.001 (0.015) [0.17]	21,508	10,227	11,281
1991	0.004 (0.003) [0.47]	0.006 (0.004) [0.36]	−0.024 (0.026) [2.60]	−0.016 (0.020) [0.55]	−0.022 (0.016) [0.17]	21,717	11,171	10,546
1992	−0.002 (0.003) [0.47]	−0.002 (0.004) [0.37]	0.042* (0.025) [2.61]	0.034* (0.020) [0.53]	0.003 (0.015) [0.17]	21,838	10,761	11,077

See notes to table 1 for the RD regression specification. Annuities are measured in year 2010 dollars. Standard errors are reported in parentheses and comparison means in brackets. * $p < 0.1$, ** $p < 0.05$.

When we look at father's long-term labor market attachment, we similarly find little evidence of an effect, either graphically for 1992 in figures 3e and 3f, or for any of the other reform years in table 3. None of the coefficient estimates are statistically significant.

The hope of many proponents of paid leave expansions is that they would promote more within-family gender equality in terms of labor force attachment. Figures 3g and 3h and table 4 reveal that the ratio of male to female employment and the ratio of male and female income annuities in the long run (fourteen years after) did not generally change in

response to the reforms. One coefficient is significant at the 10% level, but there is no obvious pattern for the remaining eleven insignificant coefficients. Since these extensions in paid leave increased the amount of time women spent at home with newborn children but did not affect men's leave taking, the reforms had the likely effect of widening the male-female child care gap, at least in the first year of a child's life.

Family structure. We finally turn to outcomes related to family structure. An additional benefit often claimed for

parental leave is that it supports and strengthens families. In Figures 3i and 3k, we look at a variety of family outcomes for the 1992 reform. The 1992 reform had a small effect on completed fertility over the next fourteen years. What about marital outcomes? Many Norwegians are not married at the birth of their child. Therefore, we can examine whether the reform increased entry into marriage or decreased exits out of marriage. We find a small effect for increased entry into marriage but no effect for divorce in 1992.

Table 4 reveals that the one marginally significant effect (at the 10% significance level) found for the 1992 reform appears to be due to sampling variability. None of the family structure estimates for the other reform years are significant, and the signs of the coefficients do not follow a consistent pattern, with a mixture of positive and negative coefficients. Our reading of the evidence as a whole is that the extensions to paid leave did little to encourage fertility or stabilize marriage.

B. Robustness

Before moving on, we present robustness results. Since there are many reforms, we show robustness for the 1992 reform; the results for other reform years are similarly robust. We begin with a series of specification checks in appendix table A3, reestimating the full set of outcomes (a) omitting control variables, (b) using separate quadratic trends on each side of the discontinuity instead of linear trends, (c) and with a one-week doughnut around the discontinuity. In each case, the results are remarkably robust and continue to point toward little effect of the reforms on our long list of outcomes.

We next examine whether our results are sensitive to different window widths. Our baseline estimates use a window of 90 days on either side of the birth cutoff date. In appendix table A4, we try a wider window of 120 days and a narrower window of 60 days. The results are similar regardless of window width, although, as expected, the standard errors increase when the window is smaller.

An alternative to using a global estimator for RD is to use local linear regression. This has the advantage of being less influenced by outliers far away from the discontinuity. We report on this final robustness check in appendix table A5. Whether we use a bandwidth of thirty or sixty days, the results are similar to our baseline estimates.

V. Costs versus Benefits

To evaluate the argument for paid parental leave, costs and benefits must be compared on a per dollar basis. We begin by discussing the costs of the program, and then translate the benefit per reform estimates from the prior section into benefits per \$1,000 spent.

A. Costs

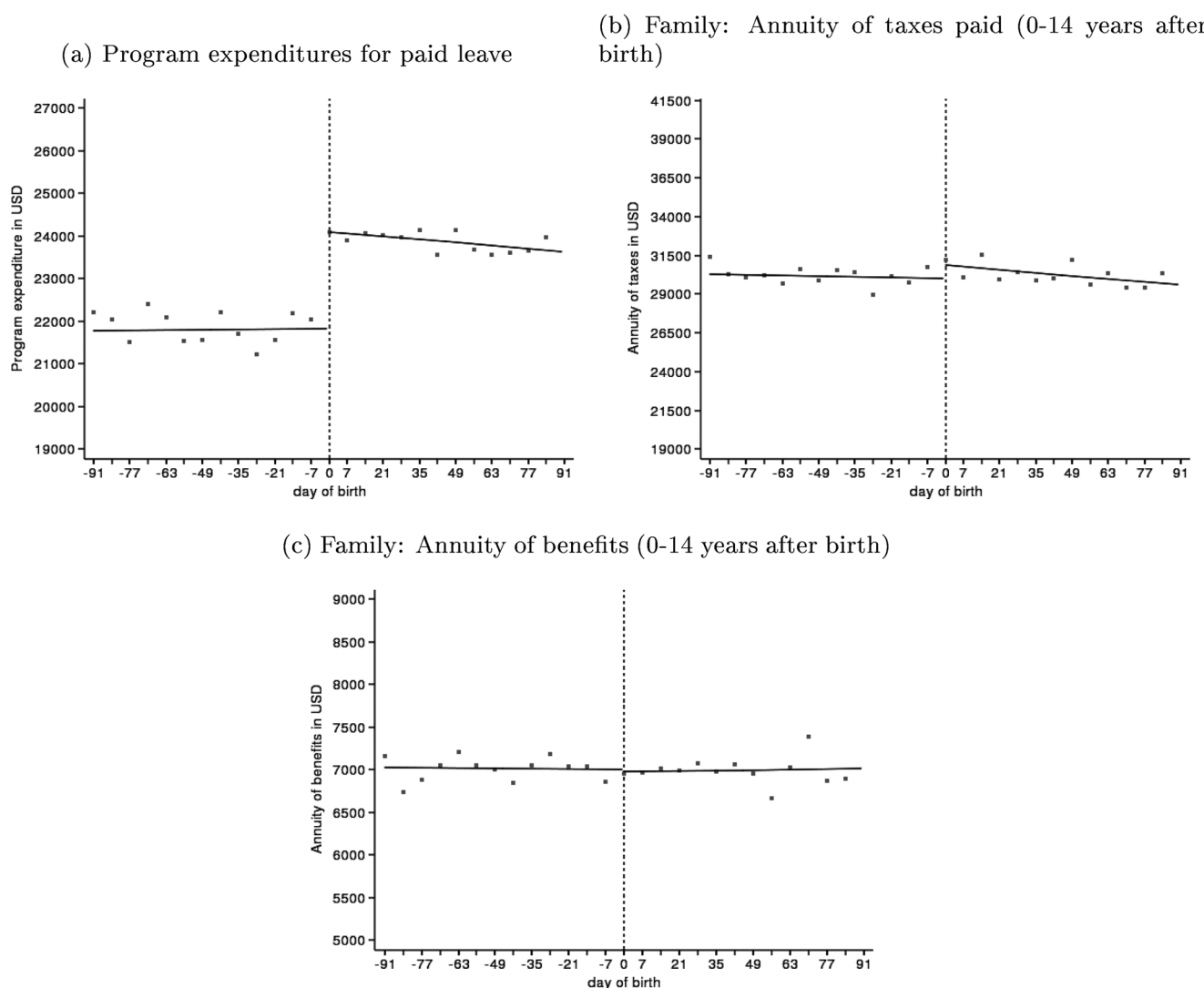
The long-run costs of paid maternity leave depend not only on direct program expenditures but also on any changes in future tax payments or participation in other transfer programs. Starting with the 1992 reform as an example, figure 4 plots the RD graphs associated with expenditures, taxes, and benefit payments per participant. As expected due to the strong take-up of the extra weeks allowed by the reform, program expenditures jump discontinuously around the reform cutoff in figure 4a. Table 5 estimates that expenditures rise by over \$2,100 per mother. Since this was a three-week reform, expenditures amount to an average transfer of roughly \$140 for each workday taken off by mothers. Similarly large and statistically significant effects are found for other reform years as well.

However, there could be indirect revenue effects from paid leave. For example, if paid leave causes mothers to work more in the future, this increase in revenues could help finance the leave program. Figure 4a graphs the annuity of taxes paid for fifteen years, starting with the year of the birth (year 0). Parental leave benefits are fully taxable at the individual level, so a portion of the leave payments is returned to the government in the form of higher tax payments in the year of the birth. However, there is also a reduction in taxes collected by the government due to the forgone earnings of the mother while on leave. Since there is 100% earnings replacement and empirically no evidence of crowd-out in unpaid leave, these two tax effects largely cancel each other out in the year of the birth. There is also little evidence of a sizable increase in tax receipt over the next fourteen years. Table 5 documents positive but generally insignificant effects on the annuity of taxes paid for the six reform years. An *F*-test reveals the estimates are not jointly significant.

Similarly, there could be future indirect costs if paid leave changes participation in other transfer programs. We find no evidence that transfer payments increase over the fifteen-year period starting with the year of the birth. This null effect is visually shown in figure 4c and verified numerically in column 3 of table 4. Hence, the total cost of the program per participant is close to the program expenditure amount, as future taxes and transfers are largely unaffected. Appendix figures A15 to A17 reveal similar expenditure, tax, and transfer patterns for the other reform years, which is perhaps not surprising given the results on family labor supply found in the previous section. Table 5 reports the estimates, which are all close to 0 and not statistically significant.

Table 5 also reports on the net costs of the program to the government, defined as program expenditures minus the annuity of taxes paid plus the annuity of benefits received. The net costs are substantial and statistically significant for four of the six reforms. The total net cost of the program depends on the number of participating mothers. In 1992, for example, 45,682 eligible women gave birth and participated in the program, at an estimated average net cost of roughly \$1,600 per mother. If a similar number of mothers continue

FIGURE 4.—COSTS, TAXES, AND BENEFITS, 1992 REFORM



Each observation is the average of the relevant outcome in a one-week bin based on the birth date of the child. The dashed vertical lines denote the reform cutoff date, which has been normalized to 0. The solid lines are estimated using a linear regression based on daily individual-level data with triangular weights. The scales of the y-axes are $\pm .5$ standard deviations of the mean outcome.

to take paid leave in the future, this implies that the yearly cost of the leave program rose by over \$75 million per year due to the three-week expansion in paid benefits.

To put the magnitude of the program in perspective, consider the cumulative net costs of the initial reform and all of the leave expansions. Based on all the reforms we study, we estimate a one-week increase in paid leave costs on net \$509 (in 2010 dollars) on average per birth. The initial 1977 reform gave mothers eighteen weeks of paid leave and one year of job protection. The series of expansions in paid leave we study in this paper expanded paid leave by seventeen weeks. Based on the number of births to eligible mothers in 1992, the annual net cost of providing 35 weeks of paid leave is \$814 million per year (in 2010 dollars). This represents almost 0.4% of GDP. Whether the initial eighteen weeks is worth the roughly \$419 million in annual costs can be debated, since Carneiro et al. (2015) find some benefits associated with the initial reform. But

the expansions nearly doubled this annual cost, without any corresponding changes in a broad range of child, family, and gender equality outcomes. These expansionary costs imply a considerable increase in taxes, at a cost to economic efficiency. While estimates of the deadweight loss due to taxes are hard to come by, Ballard, Shoven, and Whalley (1985) estimate the marginal deadweight loss to be a sizable 0.3.

B. Benefits per \$1,000 Spent

In the previous section, the potential benefits of paid parental leave were measured per reform. Some of the leave extensions were two weeks, while others were three or four weeks. In this section, we present the results for each reform in term of benefits per \$1,000 of program expenditure. These comparisons are presented graphically in appendix figure A18.

TABLE 5.—TAXES, BENEFITS, AND PROGRAM EXPENDITURES, ALL REFORMS

Reform	RD Estimates				Observations
	Program Expenditures	Annuity of Taxes Paid (years 0–14)	Annuity of Benefits Received (years 0–14)	Net Costs	
1987	1,532** (133) [11,273]	917* (533) [25,260]	45 (118) [6,570]	660 (548) [–7,417]	17,580
1988	1,257** (147) [12,805]	–61 (420) [25,808]	90 (119) [6,781]	1,408** (445) [–6,222]	19,310
1989	1,236** (153) [14,601]	511 (518) [26,750]	95 (109) [6,849]	821 (538) [–5,301]	20,093
1990	2,714** (168) [16,354]	867* (498) [27,931]	–96 (108) [6,917]	1,752** (517) [–4,660]	21,508
1991	2,797** (202) [19,512]	320 (468) [28,920]	–114 (110) [6,950]	2,363** (496) [–2,458]	21,717
1992	2,132** (219) [22,881]	450 (604) [30,240]	–38 (113) [6,999]	1,644** (630) [–360]	21,838
<i>F</i> -test	136.4	1.3	0.6	9.1	
<i>p</i> -value	.001	.253	.731	.001	

See notes to table 1 for the RD regression specification. Annuities and expenditures are measured in year 2010 dollars. Standard errors are reported in parentheses and comparison means in brackets. The *F*-test is a test of the joint significance across all six reforms in a column. * $p < 0.1$, ** $p < 0.05$.

As an example, consider appendix figure 18a, which plots the effect on academic achievement at the end of ninth grade for each reform, standardized per \$1,000 of expenditures. The *x*-axis measures weeks of paid leave, with the vertical red lines indicating each of the six extensions in paid leave. The horizontal black lines visually present the standardized estimates, while the dashed lines represent 95% confidence intervals. The graph makes clear that the reforms had little effect on children's academic achievement.

The other graphs in appendix figure 18 present similar illustrations of effect sizes per \$1,000 spent for other outcomes. One advantage of presenting results in this manner is that it is easy to compare the effect of earlier reforms to later reforms. Modest expansions when a child is still young could be worthwhile, even if later, larger expansions are not. For example, one might think the benefits of breastfeeding are more important when an infant is less than 6 months old but that this benefit fades out over time. Similarly, one might think the longer a mother stays out of the labor force, the less will be her future attachment to the labor market. However, even modest expansions, from 18 to 20 weeks or from 20 to 22 weeks, have no significant effect on outcomes. The estimates are all close to 0 and generally insignificant. Appendix table A6 presents the estimates per \$1,000 spent, averaged over all six reforms. None of the estimated average effects are large or statistically significant.¹⁶

¹⁶ There have been small changes in the composition of which mothers take leave over our sample period. Reweighting the results by observable characteristics to adjust for this does not materially affect our findings.

We next explore whether there are positive net benefits of the program to families, focusing on outcomes that can be easily converted to dollars. Our first measure of net benefits is the annuity of disposable family income, which we define as the annuity value of mother's and father's (labor income + nonlabor income + government benefits – taxes). Government benefits include maternity leave payments plus other government transfer payments. The annuity is taken from year 0 up to year 14. Our second measure adds in the high school dropout outcome for children, using an estimated return to completing high school.¹⁷

Results are presented in appendix table A6. Using the first net benefit measure, the estimate is only \$79 for a \$1,000 increase in parental leave costs. While the standard error on this estimate is relatively large, one can still reject a positive effect relative to costs, as the 95% confidence interval does not include \$1,000. Turning to the second net benefit measure, we reach a similar conclusion. There is an estimated net benefit of only \$13 for a \$1,000 increase in parental leave costs. Again, the 95% confidence interval does not include \$1,000. We conclude there is little evidence of economically significant effects, particularly relative to costs.¹⁸

At this point, it is also interesting to think about the costs versus benefits of the 1977 introduction of eighteen weeks of paid leave in Norway, as analyzed in Carneiro et al. (2015). Using estimates from their paper, per \$1,000 spent, dropout rates decline by 0.2 percentage points and children's wages increase by 0.4 percentage points. In contrast, for the additional expansions to paid leave studied in our paper, we find an opposite-signed effect (see appendix table A6) and can rule out an effect size as large as Carneiro et al. (2015) at conventional significance levels.

VI. Redistribution

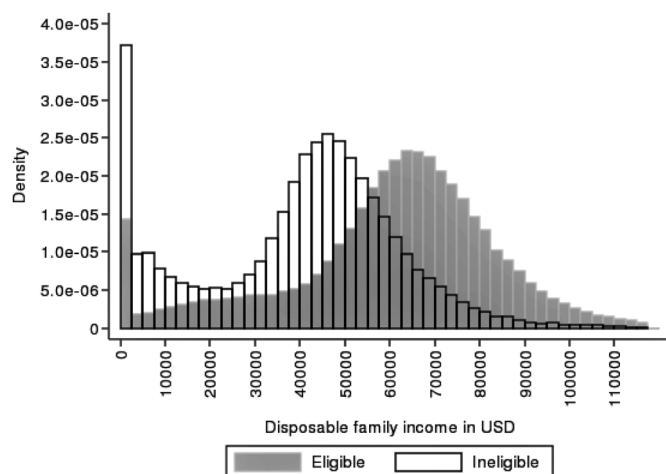
Since there are no measurable benefits associated with extensions to paid leave but large costs to taxpayers, it is important to consider the distributional effects of the program. Somewhat surprisingly, little attention has been paid in the previous literature to the redistributional effects of this social program.

As discussed previously, not all mothers are eligible for paid leave, since a requirement is that mothers have a minimum level of labor market earnings prior to the child's birth. Appendix table A7 compares eligible versus ineligible mothers on a variety of dimensions. Mothers who are eligible for parental leave benefits are more educated, are married

¹⁷ To calculate the return to completing high school, we use Norwegian panel data with nearly career-long earnings histories (see Bhuller, Mogstad, & Salvanes, 2014). We estimate the impact of dropping out of high school on the discounted value of lifetime earnings, controlling for municipality, municipality of birth, and cohort fixed effects. We use this to predict the earnings effect of the paternity leave-induced change in high school completion.

¹⁸ An important caveat is that these calculations do not include nonmonetary benefits; in particular they do not include the utility that mothers get from increased leisure time.

FIGURE 5.—DISTRIBUTION OF DISPOSABLE FAMILY INCOME THE YEAR PRIOR TO BIRTH, FOR ELIGIBLE AND INELIGIBLE MOTHERS



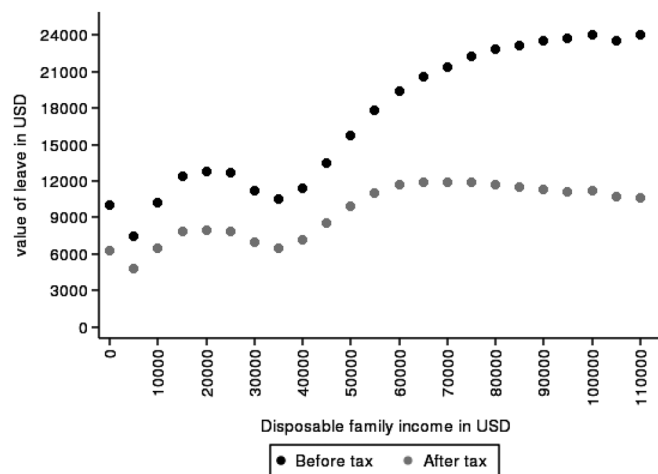
Sample of eligible and ineligible mothers aged 25 to 40 giving birth during the period 1987 to 1992. Disposable income is calculated as the sum of parents' combined labor income, parental leave benefits, nontaxable transfers, unemployment benefits, and sick leave minus taxes paid and is measured the year before birth and converted into year 2010 dollars. The top 1 percentile of disposable family income is excluded from the graph. Each bin in the histogram is \$2,500 wide.

to higher-educated men, and are more likely to be natives compared to ineligible. Eligible mothers also have substantially higher family income compared to ineligible mothers, with disposable income being 60% higher in the year prior to birth. This income gap persists over time, with family income for eligibles being almost 20% higher on average compared to ineligible mothers in 2010.

The distributions of family income for eligible and ineligible mothers are presented in figure 5. The graph plots the distributions for family income in the year prior to the child's birth. The graph makes clear that paid parental leave makes transfers to relatively well-off mothers. Not only is the mean of the eligible distribution further to the right, but the left tail of the income distribution has a much larger mass for ineligible. Twenty-five percent of the ineligible distribution has family income below \$25,000 compared to only 10% of the eligible distribution. In addition, only 16% of the ineligible distribution has family income above \$60,000 compared to 58% of the eligible distribution. These are large distributional disparities.

Even within the group of eligible mothers, the parental leave program transfers larger amounts to families with higher earnings. Leave benefits are tied to prior earnings, and women with higher earnings reside in families with higher income. Since the program gave 100% income replacement up to a high threshold, the amount of the transfer tends to increase with family income. The black dots in figure 6 plot the pretax value of the leave transfer to mothers versus disposable income, defined as the sum of parents' labor income plus taxes and transfers. Both eligible and ineligible mothers are included in the graph. The pretax transfer is much larger for higher-income families: families earning less than \$25,000 receive \$10,560 in paid leave on average, while

FIGURE 6.—AVERAGE VALUE OF PAID LEAVE TRANSFER BY DISPOSABLE FAMILY INCOME, BEFORE AND AFTER TAX



Sample of all mothers who gave birth between 1987 and 1992. Disposable income is defined in the note to figure 5. The top 1 percentile of disposable family income is excluded from the graph. Each observation in the graph plots the average value of the leave transfer for a \$5,000 interval.

families earning greater than \$60,000 receive \$21,349 on average. A considerable part of this disparity is undone by Norway's progressive tax system, as leave benefits are fully taxable at the individual level. But as the gray dots in figure 6 show, even after taxes, the program continues to favor middle- and upper-income families, with families earning more than \$60,000 receiving roughly 75% more in after-tax leave benefits compared to families earning less than \$25,000.

The differences in income between ineligible and eligibles are even larger when comparing eligible mothers to the relatively poor men and women who never have children. The second panel of appendix table A7 documents that family income in 2010 is almost two and a half times larger in eligible families compared to childless individuals, although it should be noted that eligible families have larger household sizes compared to those who never have children.

The comparisons in this section make clear that paid parental leave has negative redistribution properties. The program makes regressive transfers from both ineligible mothers and childless individuals to eligible mothers. This is because ineligible mothers and childless individuals have lower incomes on average but receive nothing from the program. Since there is no crowd-out of unpaid leave, the extra leave payments amount to a pure leisure transfer to middle- and upper-income families, which are eligible at the expense of some of the least well off in society. Within the group of eligible mothers, the program makes larger payments to families with higher incomes. The transfer is proportional in eligible mothers' earnings, since there is 100% earnings replacement and progressive in after-tax benefits among eligibles (since the transfer is taxable and the tax schedule is progressive). Of course, how the program is financed and whether other publicly funded transfer programs are

crowded out ultimately determine the net distributional effect of maternity leave payments. For example, if expansions to maternity leave crowd out spending on social safety net programs for the poor, then maternity leave makes the combined tax and transfer system less progressive. How the funds for this program were raised and whether maternity leave crowded out spending on other programs are difficult questions to answer and beyond the scope of this paper.

VII. Conclusion

Paid parental leave has gained greater salience in the past few decades as mothers have increasingly entered the workforce. Indeed, the median number of weeks of paid leave among OECD countries was 14 in 1980 but had risen to 42 by 2011. The usual arguments for paid leave are that the program addresses important social goals or market failures. However, empirical evidence on the benefits and costs of paid leave is scarce, and many key questions are still unanswered. First, does paid leave increase available parental time with children, or does it simply crowd out unpaid leave? Second, what effect does paid leave have on a broad range of child, parent, and family outcomes? Third, how do any benefits compare relative to costs? And finally, are there any distributional effects?

We answer these questions in the context of Norway's parental leave program, focusing on parents' responses to a series of policy reforms that expanded paid leave from 18 to 35 weeks (without changing the length of job protection). Our first empirical result is that none of the reforms seem to crowd out unpaid leave. Since income replacement was 100% for most women, the reforms caused an increase in mother's time spent at home after birth, without a reduction in family income. This allows for a clean estimate of the effect of parental time on child and family outcomes, as there are no income effects to worry about. Second, the expansions had little effect on a wide variety of outcomes, including children's school outcomes, parental earnings, and participation in the labor market in the short or long run, completed fertility, marriage, or divorce. Third, paid parental leave has negative redistribution properties: it makes regressive transfers from ineligible mothers and childless individuals to eligibles, and among the group of eligible mothers, the payments increase with family income. Since there was no crowd-out of unpaid leave, the extra leave benefits amounted to a pure leisure transfer. Finally, the reforms extending paid leave had little impact on parents' future tax payments and benefit receipt. As a result, the large increases in public spending on parental leave imply a considerable increase in taxes, at a cost to economic efficiency.

Taken together, our findings suggest the case for extensions to paid maternity leave must rest on arguments other than improvements in observable dimensions of public benefits, redistribution, or market failures due to credit constraints. In a time of harsh budget realities, our findings are particularly relevant for countries that are considering future expansions

or contractions in the duration of paid leave. One can argue whether a baseline level of paid parental leave or job protection is worth the cost. But the case for extended paid leave periods, which are prevalent in OECD countries, is weak. One would have to argue there are benefits not captured in the relatively rich set of outcome variables we look at; for example, it is possible that extended parental leave could result in improved health outcomes for mothers and their children. Additionally, our empirical strategy identifies only the responses of parents with children born around the time of the reforms, which might not be externally valid for parents having children in later periods. In a long-run perspective, when firms might adjust to the extensions of parental leave and women might change their investments in human capital and fertility behavior, the effects of the program could potentially be stronger. Unfortunately, costs and benefits of such long-run adjustments, if they exist, are difficult to verify. By way of comparison, the opportunity costs to funding parental leave are substantial and verifiable: a growing body of evidence shows that government subsidies to early education can yield high returns in the short and long run, particularly for disadvantaged children.¹⁹

¹⁹ See, for example, the review of this literature by Almond and Currie (2011).

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